

6.6.6.3. Derivation of ISD from Incoming Message

The algorithm to derive the ISD from an incoming message takes as input:

- The incoming message (**message**)
- The Session ID of the incoming message (**session_id**)
- A conceptual Sessions Metadata database (**sessions_metadata**)
- The Group Key Management Cluster (**group_key_management_cluster**)

The output of the algorithm is the **SubjectDescriptor** structure below:

```

DEFAULT_COMMISSIONING_PASSCODE = 0

enum AuthModeEnum {
    None = 0, # conceptual "no auth" value
    PASE = 1,
    CASE = 2,
    Group = 3
}

struct SubjectDescriptor {
    bool IsCommissioning;
    AuthModeEnum AuthMode;
    list<SubjectID> Subjects; # max 3 items
    FabricIndex FabricIndex;
}

```

The algorithm SHALL function as follows:

```

def get_isd_from_message(message) -> SubjectDescriptor:
    isd = {
        IsCommissioning: False,
        AuthMode: AuthModeEnum.None,
        Subjects: [],
        FabricIndex: 0
    }

    session_id = message.get_session_id()

    if sessions_metadata.get_auth_mode(session_id) == AuthModeEnum.PASE:
        isd.AuthMode = AuthModeEnum.PASE
        isd.IsCommissioning = True
        isd.Subjects.append(DEFAULT_COMMISSIONING_PASSCODE)

```